

Tox21 Molecular Toxicity Prediction Using an MLOps Deployment Pipeline

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Abstract

This report presents an end-to-end MLOps pipeline for molecular toxicity prediction using the Tox21 dataset. The project converts molecular SMILES strings into 1024-bit Morgan fingerprints with RDKit, trains and compares Logistic Regression, Random Forest, and Extra Trees classifiers, tracks experiments with MLflow, serves the selected model through FastAPI, and exposes operational metrics through Prometheus and Grafana.

Keywords

MLOps, Tox21, molecular toxicity prediction, FastAPI, Docker, MLflow, Prometheus, Grafana, GitHub Actions.

I. Introduction

This project focuses on a reproducible MLOps workflow for the Tox21 SR-ARE toxicity endpoint. The repository includes the dataset, training pipeline, selected model artifact, FastAPI inference service, Docker configuration, Prometheus metrics, Grafana dashboard provisioning, CI validation, and CD image publishing workflow.

II. System Architecture

High-Level Architecture of the Tox21 MLOps Pipeline for Molecular Toxicity Prediction

Current Orchestration

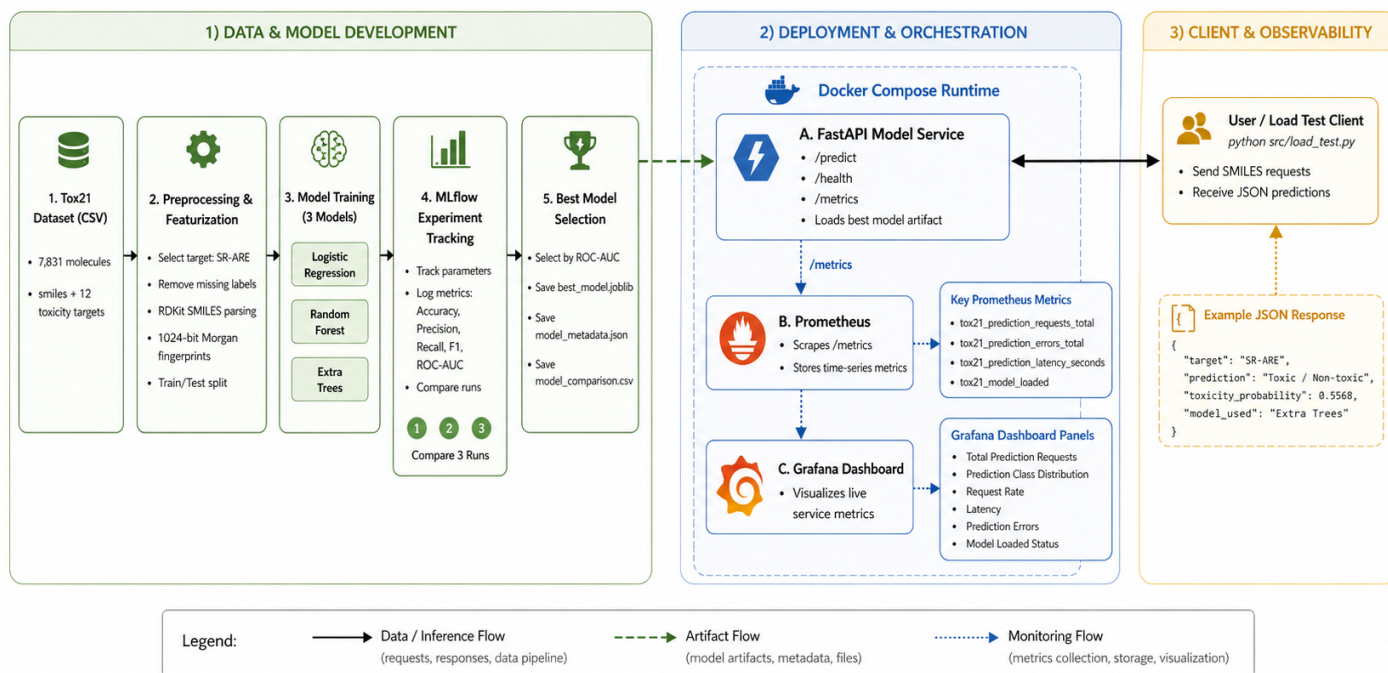


Fig. 1. High-level architecture of the Tox21 MLOps pipeline for molecular toxicity prediction.

The architecture is organized around data and model development, Docker-based deployment and orchestration, and client observability. The API exposes prediction, health, version, and metrics endpoints.

III. Dataset and Feature Engineering

Item	Value
Dataset	Tox21
Target	SR-ARE
Input	SMILES string
Feature type	Morgan fingerprints
Fingerprint size	1024 bits
Fingerprint radius	2

The training script removes records with missing target labels and skips molecules that RDKit cannot parse. The same fingerprint representation is used during model training and API inference.

IV. Model Training and Selection

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7494	0.3443	0.6117	0.4406	0.7443
Random Forest	0.7923	0.3846	0.4787	0.4265	0.7592
Extra Trees	0.7700	0.3571	0.5319	0.4274	0.7648

Extra Trees is selected as the best model using ROC-AUC as the selection metric. The selected artifact is stored at models/tox21_best_model.joblib, with metadata stored in models/model_metadata.json.

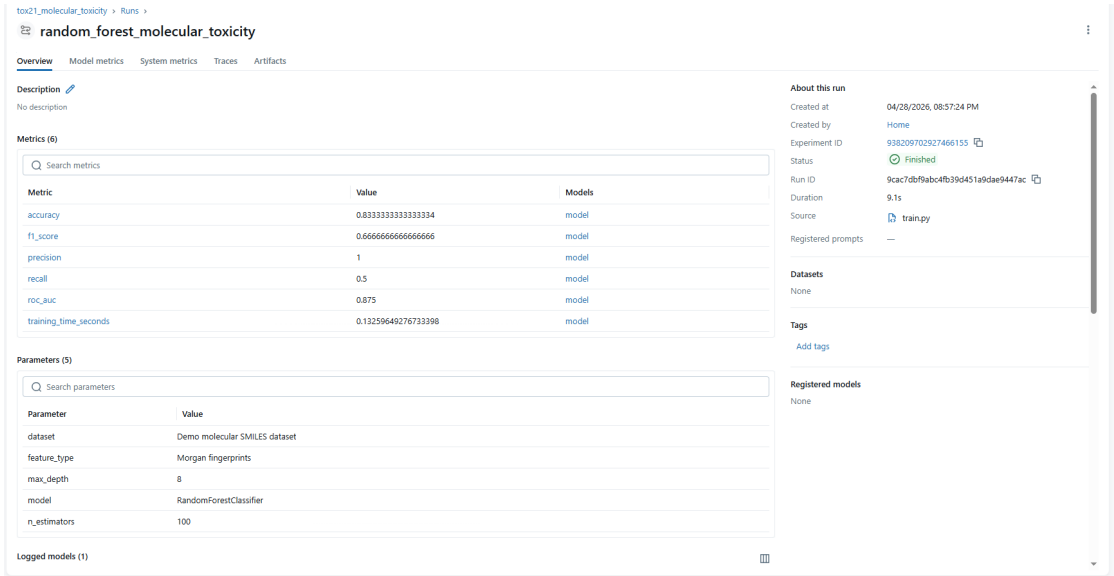


Fig. 2. MLflow run comparison captured from the project artifacts.

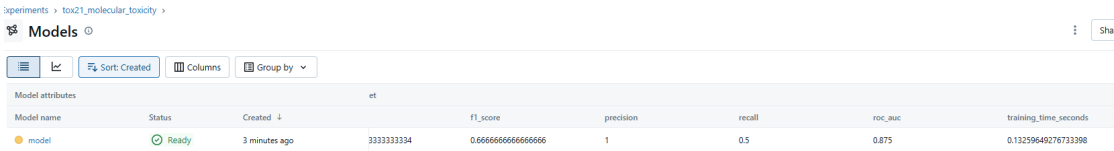


Fig. 3. Model evaluation screenshot captured from the project artifacts.

V. API Serving Layer

Endpoint	Purpose
GET /	Service root message
GET /health	Model and service health metadata
GET /version	App, model, and runtime version metadata
POST /predict	Toxicity prediction for a SMILES input
GET /metrics	Prometheus-compatible service metrics

VI. Observability

The API exposes Prometheus metrics for prediction attempts, outcomes, errors, response status codes, latency, and model load status. Grafana provisioning is included in the repository and groups panels around service health, traffic quality, and latency.

VII. CI/CD and Reproducibility

The CI workflow installs pinned dependencies, verifies required files, compiles Python files, runs tests, checks model artifact compatibility, and builds the Docker image on pushes. The CD workflow publishes the API image to GitHub Container Registry and supports semantic version tags.

VIII. Limitations

The project is intended as an MLOps demonstration, not a validated toxicology decision system. The model is specific to the SR-ARE endpoint and uses fixed fingerprint features. Real-world safety-critical use would require stronger validation, calibration analysis, data governance, monitoring, and domain expert review.

IX. Conclusion

The project demonstrates a complete MLOps workflow for molecular toxicity prediction: preprocessing, feature generation, model comparison, experiment tracking, artifact management, API serving, containerization, observability, CI validation, and image publishing.

Missing Information To Complete Final Submission

TODO: author name, institution or department, course or module name, instructor or supervisor name, submission date, and required citation style.